

LAYER 3 OF 13

DATABASE STORAGE

Where Your Data Lives

TIER 1 — SOLOPRENEUR

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Part of the Builder Certification Series

What This Kit Covers

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This kit covers Layer 3: Database & Storage—where your data lives. Every app needs to remember things. User accounts, orders, messages, uploaded photos, settings, transaction history—all of that has to be saved somewhere so it's still there when someone comes back tomorrow. That somewhere is your database and storage system.

You're not going to set up databases by hand. You're going to describe what your app needs to remember—"I need to store customers, their orders, and photos of their completed work"—and let your AI coding tool build the data layer for you. Your job is to understand how data is organized, check that things are being saved correctly, and make sure nothing important gets lost. That's what this certification proves you can do.

This kit walks you through three levels of mastery. Tier 1 is for someone building their first app who needs data to save and load reliably. Tier 2 is for someone whose app is growing and needs to keep data organized, fast, and backed up. Tier 3 is for someone running a platform at scale where data loss or downtime is unacceptable. Start at Tier 1. You can always come back for the rest.

Why It Matters

If your frontend is the face and your backend is the brain, your database is the memory. Without it, your app forgets everything the moment someone closes it. Imagine a to-do app that loses all your tasks overnight, or an e-commerce store that forgets what you ordered. A broken or disorganized database doesn't just cause bugs—it destroys trust.

Here's the thing about AI coding tools: they'll happily set up a database for you, but they often make choices you didn't think about—like storing data in a way that gets slow as your app grows, not setting up backups, or creating duplicate records because there's no rule preventing it. AI builds the structure you describe. If you don't know what good data organization looks like, you get a mess that's painful to untangle later. This kit teaches you what good looks like.

Certification Tier Map

There are three certification tiers. Each one builds on the one before it. Here's a quick look at who each tier is for and what you'll prove you can do.

Tier

Who It's For

What You Prove

Tier 1

Solo builders shipping their first app

You understand how data is stored, can describe your data structure to AI, and can verify that data saves and loads correctly.

Tier 2

Growing teams with more data and users

You can manage database changes safely, optimize for speed, handle file storage, and keep your data backed up and recoverable.

Tier 3

Platforms with high volume and compliance needs

You can oversee database scaling, replication, data warehousing, and compliance-grade data governance at enterprise scale.

TIER 1 CERTIFICATION GOAL

You can describe what data your app needs to store, use an AI coding tool to set up the database, verify that data saves and loads correctly, and ship an app where nothing important gets lost.

What You Need to Know

You don't need to write database queries by hand. You need to understand a handful of concepts so you can describe your data clearly and catch problems in what AI gives you back.

What a database actually is: A database is a structured place to store information—like a spreadsheet, but built for apps. Instead of opening a file and scrolling through rows, your app sends requests to the database to save, find, update, or delete information. The most common type you'll see is a relational database (like PostgreSQL), which stores data in tables with rows and columns—just like a spreadsheet with strict rules about what goes where.

What a schema is (your data's blueprint): A schema defines the structure of your data—what tables exist, what columns each table has, and what type of data each column holds. For example, a "users" table might have columns for name (text), email (text), and created_date (date). Think of it as the blueprint your database follows. If you don't define a clear schema, AI will make one up—and it might not match what your app actually needs.

SQL vs. NoSQL (two styles of database): SQL databases (like PostgreSQL, MySQL) organize data in structured tables with defined relationships—great for apps where data is predictable, like orders linked to customers. NoSQL databases (like MongoDB, Firebase) are more flexible and store data as documents—like filing folders that can hold anything. For most apps you'll build with AI, SQL is the default and the right choice.

How tables relate to each other: In most apps, data is connected. A customer has many orders. An order has many items. These relationships are defined in your schema. If AI doesn't set up relationships correctly, you'll end up with orphaned data—orders that don't connect to any customer, or items floating without an order.

The vibecoder workflow for databases: Describe what your app needs to remember → AI sets up the database and schema → you check that data saves correctly by creating test records → you verify that loading data returns the right results → you fix any gaps → ship it. Same loop as the other layers, but you're checking that data goes in right and comes out right.

Your Toolkit

Your AI coding tool sets up the database. These tools help you inspect and verify it.

AI coding tool (pick one): Cursor, Lovable, Bolt, Claude Code, Windsurf, or whatever AI-assisted builder you prefer. This is where you'll describe your data structure and let AI build the database layer.

A database with a visual interface: Supabase or PlanetScale give you a web dashboard where you can see your tables, browse rows, and run queries without touching code. Think of it as looking at your spreadsheet directly instead of trusting that it's correct.

A database viewer: Tools like TablePlus or the Supabase dashboard let you browse your data visually—see what's actually stored, check for duplicates, and verify relationships between tables.

Certification Exam Topics

Every exam question is scenario-based. You'll see a situation and need to identify what's right, what's wrong, or what to do next. Here's what gets tested:

Schema design: Can you evaluate whether AI created the right tables and columns for your app's data—or missed something important?

Data types: Can you identify when AI used the wrong type for a column—like storing a price as text instead of a number, or a date as a plain string?

Relationships: Can you spot when tables that should be connected aren't—like orders that don't link back to the customer who placed them?

Required vs. optional fields: Can you tell when a column should require a value (like an email for a user account) vs. when it should allow empty values?

Duplicate prevention: Can you identify when your database allows duplicate records that shouldn't exist—like two accounts with the same email address?

Basic queries: Can you describe what data you need and verify that the results are correct—even if you're not writing the query yourself?

Data integrity: Can you check that deleting a record doesn't leave orphaned data behind—like deleting a customer but leaving their orders in the system?

Your AI workflow: Can you describe a data structure clearly, inspect the result in a database viewer, give AI useful feedback when something is wrong, and iterate to a correct schema?

Common Pitfalls

These are the mistakes vibecoders make most often at this layer. No judgment—they're easy to make. But if you recognize any of them in your own workflow, fix them before sitting for the exam.

Never looking at your actual data. You trust that AI set up the database correctly, but you've never opened a database viewer to check. There could be duplicate records, missing relationships, or columns with the wrong data types—and you'd never know until something breaks in production.

No backups. Or worse—you have backups but you've never tested restoring one. A backup you've never restored is just a file that makes you feel better. Test it before you need it.

Storing files directly in the database instead of a file storage service. AI will do this if you don't specify otherwise. Your database will get huge and slow. Photos, PDFs, and uploads belong in file storage (like Supabase Storage or S3), with just a link stored in the database.

Not thinking about what happens when your data grows. Your app works great with 50 test records. But with 50,000 real records and no indexes, that search query takes 30 seconds. Ask AI to add indexes on any column you'll search or filter by.

Letting AI create tables without unique constraints. If there's no rule saying "each email can only exist once," your users table will eventually have duplicate accounts, and you'll spend days untangling the mess.

Making schema changes directly on the production database. One wrong move and you lose real user data. Always use migrations—planned, tested changes that can be rolled back if something goes wrong.

Tier 1 Self-Assessment Checklist

Before you take the exam, run through these questions. Every "no" is something to work on.

Can you describe your app's data structure to AI clearly enough that it creates the right tables and relationships?

Have you actually opened a database viewer and looked at the data AI created—checking for correct types, relationships, and constraints?

Does your database prevent duplicate records where they shouldn't exist (like two accounts with the same email)?

If you deleted your database right now, could you restore it from a backup?

Are uploaded files (photos, documents) stored in a file storage service—not directly in the database?

Can you describe what happens to related data when a record is deleted (like what happens to orders when a customer is removed)?

Do you know how to check whether your database queries are fast enough—and what to do if they're not?

AI Audit Prompt Template

Copy this prompt into your AI coding tool to get a quick health check on your database. It checks the same things the certification exam covers.

Review my app's database and storage setup and check the following. For each one, tell me pass or fail with a specific example:

Schema design: Are my tables logically organized with appropriate data types, required fields, and clear naming conventions?

Relationships: Are related tables properly connected with foreign keys—and are there rules for what happens when a parent record is deleted?

Unique constraints: Are there constraints preventing duplicate records where duplicates shouldn't exist (like duplicate email addresses or duplicate order numbers)?

Indexes: Are commonly searched and filtered columns indexed for fast lookups—especially as data grows?

File storage: Are uploaded files (images, documents, videos) stored in a proper file storage service instead of directly in the database?

Backups: Is there an automated backup system in place, and has a restore been tested successfully?

Give me an overall score out of 6 and list the top 3 things to fix first.

What's Next

Once you've gone through this kit and can answer "yes" to the self-assessment checklist, you're ready for the Layer 3 certification exam at your target tier.

The best way to prepare: build something real. Pick a project that needs to store data—even a simple one like a task tracker or a customer directory—and go through the full loop. Describe your data to AI. Check the schema it creates. Look at the actual data in a viewer. Test what happens when you delete records. Fix the gaps. Ship it. Every mistake you catch during building is exactly what the exam tests.

CERTIFICATION PATHWAY

Associate Builder: Pass all 13 layers at Tier 1

Certified Builder: Pass all 13 layers at Tier 1 + Tier 2

MADE Certified: Pass all 39 tier exams + capstone project

Each exam requires 80% to pass. You can retake after a 24-hour cooldown. No rush—take the time to build something real first.

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